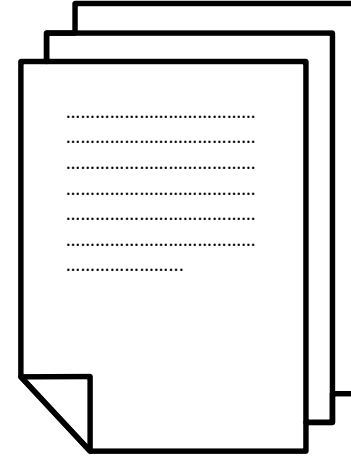


Documents

○ Document = the **element** to be retrieved

- Unstructured nature
- Unique ID
- N documents --> **Collection**



- web-pages, emails, book, page, sentence, tweets
- photos, videos, musical pieces, code
- answers to questions
- product descriptions, advertisements
- people

Queries

- Free text to express user's information need
- Same information need can be described by different queries
 - Are chatting Apps secure?
 - Live chat protection
 - Breaches in online chat
- Same query can represent different information needs
 - Apple
 - Jaguar



Queries – different forms

- Web search → keywords, narrative ...
- Image search → keywords, sample image
- QA → question
- Music search → humming a tune
- Filtering/recommendation → user's interest/history
- Scholar search → structured (author, title ..)
- Advanced search
 - `#wsyn(0.9 #field (title, #phrase (homer,simpson)) 0.7 #and (#> (pagerank,3), #ow3 (homer,simpson)) 0.4 #passage (homer, simpson, dan, castellaneta))`

Relevance

- At an abstract level, IR is about:
 - does item ***d*** match query ***q***? ... or ...
 - is item ***d*** relevant to query ***q***?
- Relevance is a tricky notion
 - will the user like it / click on it?
 - will it help the user achieve a task? (satisfy information need)
 - is it novel (not redundant)?
- Relevance → similarity
 - i.e. ***d***, ***q*** share similar “meaning”
 - about the same topic / subject / issue

Information Need/Query/Relevance

○ Information need

- Topic about which the user desires to know more
- In the user's mind!

○ Query

- What the user conveys to the computer
- Considered one representation of the information need

○ Relevance

- Document having a value with respect to the information need
 - i.e., a document is relevant if it satisfies the information need